Electrical Engineering is not an easy field to get into. These engineers must combine difficult mathematics with applied sciences and must maintain most of today's technologies while maintaining ethical duties. At Iowa State University, I had to face many challenges in all my courses that were in different technical fields as well as taking on ethical dilemmas for technological purposes. The University taught me how to look at a problem, address the issue, make it practical, and find a solution. This was done in many facets where to incorporate my own critical thinking, design experience, teamwork analysis, and ethics to get solution to whatever issue I am dealing with. In this paper we will see how my skills and background came into play to help me on this journey through college.

When it came to course work at the University, Circuit courses train the students at a very incredible level. In Circuit Analysis courses, students are tasked with understanding how to build systems that can have a purpose and working with other students to get a project done. I had to experience this by designing systems such as a temperature sensor, a system that relates voltage to temperature in EE 2300 and eventually a more complex design of it in EE3300 where I learned more at the transistor level what is happening to voltage and current with how it relates to temperature. By applying my research and development skills I gained by facilitating projects in courses I have taken over the years and knowledge I learned as an undergraduate researcher, I was able to get the task done successfully. I did this by finding papers on the issue at hand to learn more about it, coming up with ideas of my own that I can explore to solve the issue, find what works and what doesn't, design a schematic of it, build the system, validate and test it, and solve the issue. I did not do all of this in my courses alone; I did this with friends I made along the way. As a new student at college, you start at ground zero. You are not facing the struggles of life that everyone faces in high school, everyone is equal and discusses with each other. When labs start students are put into groups and collaborate to solve the problems of the lab just like a real engineering job would have. I worked in a team with many people from many diverse backgrounds and was able to learn more about their views on how to solve a problem and I was able to learn from them because of it. I was able to increase my skills of working in a technical field and I am most certainly appreciative of Iowa State for that reasoning. Because of this opportunity I was able to meet people from diverse backgrounds and was able to increase my communication skills by understanding the language of others and how they facilitate their operations so that I can understand everyone more. I was able to gain a higher level of proficiency in communication amongst teams for technical projects because I tried to take on a leadership role and learn more about the others in my teams. However, it is not always about how well you can get along with your team. At Iowa state one thing that new students learn about later is ethical dilemmas of what we do, this is a

major topic that Iowa State University brought in that has better shaped engineers. Ethics is a battle of what is considered right and wrong and with engineering this can be hard to figure out. The reasoning is that sometimes engineers think of something scientifically related and believe they have the power to make this new revelation, however some people may find the revelation as immoral, so companies and engineers have to consider ethics as a standard part of the process to maintain equality of a solution. I faced this in my projects at school. Sometimes I was allowed to use AI to help assist me in projects like the temperature sensor., but the ethical background comes into effect when a student poses the question of how much Ai is allowed to use before it is considered cheating or malpractice. For me Ethics has been a serious topic in engineering because the problems we experience as engineers are difficult and our results can be hard to make people believe in to be right. In most fields of electrical engineering, what electrical engineer is designing influences many people. Most people in the world can be affected by engineers' designs in many ways. For example, an engineer can build a High Voltage system that if done wrong and people nearby can be fatal for citizens nearby. Something like this can have serious effects globally, economically, and environmentally. Engineers can mess with the economy by designing something that causes people to lose trust in a company that is well known in a country and then cause ta countries economy to lower. It can hit globally if the design is brought outside, it is country of origin and was a bad design. Depending on the project it can be bad for the environment and can affect vast amounts of land. For example, if a power company adds more power lines, they would have to disrupt the soil where the power line will go. I experienced these hands on in my internships as an engineer I worked outside in the field for construction job. In these jobs construction workers had to dig up roads, reroute traffic and create loud environments for the people who lived nearby. I designed parts of the project that were being installed into the construction of the new road. By contributing to the design and overseeing the project happening I have to deal with these dilemmas discussed above. Economically, by my organization rerouting traffic can cause people to stay at home to not deal with the construction which can affect the town's economy since less GDP is being created. Environmentally I had to deal with Environmental Engineers from the town who came in to make sure we met EPA rules. For Global, I dealt with teams from other countries like India who were working on the project, which in return effects India since representatives of that country were a part of the matter. For Societal concerns, most people do not like it when construction starts because it effects normal driving conditions and thus can create angry local citizens to shop up and complain. This creates ethical challenges for an Engineer because some people may like their decisions but other may dislike it and the design can be damaging. So, engineers really must think through the best possible decision technically and ethically. This is a struggle for not only me but for all engineers in the world nowadays because everyone

comes from a different background and thus everyone has a different viewpoint which in return creates different ethic practices for everyone.

Although this seems difficult in many ways, the education here at Iowa State can foster my decision making for ethical and technical dilemmas. As students in Electrical Engineering, we must take many rigorous courses in fields like Electronics, Embedded Systems, Computer Engineering, Power Engineering, Electromagnetism, Signal Processing, and much more. For many it is hard to see the overlap in education because they all seem like vastly different subjects. However, if students sit down and think about it, they are all related to each other in some form. This means whatever I learn in one class I can harness that knowledge and use it for another course in a different but related field. This is a way of thinking that few people think about. This can be especially useful because it allows students to stop and think creatively when dealing with a difficult problem. At the end of the day Iowa State University is giving students like me a degree in Electrical Engineering but what they are doing is molding us into professional analytical people. They are doing this by encouraging students to work on difficult homework problems, labs, work in teams, give projects, etc. By doing this it allows the brain to develop critical thinking skills and allows students to become good engineers. The ethical background is molded by your personal background and what you learn through the University. By learning about the rights in wrongs in the electrical world then students going into full-time positions can stop and think about ethical dilemmas a lot more easily compared to others who have not had to deal with ethics before.

When I am done with college all what Iowa State University has taught me will stick with me for years to come. The lessons taught will help me get my first job, create an environment to test the technical knowledge I learned as well as witness live ethical dilemmas on display and will create an atmosphere of new learning. Longterm this will affect my life in ways that can only benefit me. Iowa State taught the surface material I needed to start my journey but, in the end, I will have wise experience in my technical focus and will be able to clearly identify the ethical issues at hand and address them accordingly. As a final note, when I get out of college I will also still teach myself new skills in EE that deal with ethics and technical information so I stay up to date with new technological needs.